INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

End Semester Examination, December - 2024

Degree: **B.C.A**, Semester: **Third**

Stream: **BCA**

Paper Code: **BCAMD301**

Paper Name: **Computer Networks**

Full Marks:100 Duration:3 Hours

Part - A

Attempt 10 questions out of 15 questions

Each question carries 2 Marks (2 X 10)

1. Identify the five components of a data communications system. **2**

2. What are the two types of line configuration? **2**

3. For n devices in a network, what is the number of cable links required for a mesh, ring, bus and star topology? **2**

4. Why are protocols needed? **2**

5. Which layers in the Internet model are the network support layers? **2**

6. Name some service provided by the application layer in then Internet model. **2**

7. Distinguish between data rate and signal rate. **2**

8. Define carrier signal and its role in analog transmission. **2**

9. What are the three major classes of guided media? **2**

10. How does a single-bit error differ from a burst error? 2

11. How is a repeater different from an amplifier? 2

12. Find the class of the following IP addresses. 2

    a) 208.34.54.12
    b) 238.34.2.1
    c) 114.34.2.8
    d) 129.14.6.8

13. What is the minimum size of a UDP datagram? 2

14. What is the maximum size of the process data that can be encapsulated in a UDP datagram? 2

15. What is the difference between a primary server and a secondary server? 2

Part - B
Attempt 6 questions out of 9 questions
Each question carries 5 Marks (5 X 6)

16. What are the advantages of distributed processing? 5

17. How does information get passed from one layer to the next in the Internet model? 5

18. What are the responsibilities of the data link layer in the Internet model? 5

19. What is the difference between a port address, a logical address, and a physical address? 5

20. What is the difference between a bus backbone and a star backbone? 5

21. What is dotted decimal notation in IPv4 addressing? What is the number of bytes in an IPv4 address represented in dotted decimal notation? What is hexadecimal notation in IPv6 addressing? What is the number of digits in an IPv6 address represented in hexadecimal notation? 5

22. What is a mask in IPv4 addressing? What is a default mask in IPv4 addressing? 5

23. Change the following IPv4 addresses from binary notation to dotted-decimal notation. 5

    a) 10000001 00001011 00001011 11101111
    b) 11000001 10000011 00011011 11111111

24. Change the following IP addresses from dotted-decimal notation to binary notation. 5

    a) 114.34.2.8
    b) 129.14.6.8
    c) 208.34.54.12
    d) 238.34.2.1

Part - C
Attempt 5 questions out of 8 questions
Each question carries 10 Marks (10 X 5)

25. What is the bandwidth of a signal that can be decomposed into five sine waves with frequencies at 0, 20, 50, 100, and 200H_z? all peak amplitudes are the same. Draw the bandwidth. 10

26. Draw the graph of the NRZ-L scheme using each of the following data streams, assuming that the last signa 1 level has been positive. From the graphs, guess the bandwidth for this scheme using the average number of changes in the signal level. 10

    a) 00000000
    b) 11111111
    c) 01010101

d) 00110011

27. In CRC, show the relationship between the following entities (size means the number of bits): **10**

    a) The size of the data dataword and the size of the codeword.
    b) The size of the divisor and the remainder.
    c) The degree of the polynomial generator and the size of the divisor.
    d) The degree of the polynomial generator and the size of the remainder.

28. Define controlled access and list three protocols in this category.

29. Compare and contrast a random-access protocol with a channelizing protocol. **10**

30. Explain why most of the addresses in class A are wasted. Explain why a medium-size or large-size corporation does not want a block of class C addresses. **10**

31. Find the range of addresses in the following blocks. **10**

    a) 123.56.77.32/29
    b) 200.17.21.128/27
    c) 17.34.16.0/23
    d) 180.34.64.64/30

32. Compare the TCP header and the UDP header. List the fields in the TCP header that are missing from UDP header. Give the reason for their absence. **10**

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